

## 5.5.1 Functions of Equity and Equal Access

انصاف اور مساوی رسائی کے چار کام ہیں: ضرورت کے مطابق مدد (جیسے نابینا طالب علم کے لیے آڈیو وضاحت)، رکاوٹیں ہٹانا...

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 14 of 15

★ Golden

### 5.5.1 Functions of Equity and Equal Access ★ Golden

#### LEARN IT

Equity and equal access are not slogans. They **do work** in a digital classroom. The source lists four functions.

| Function                                        | What it does                                         | Pakistani example                                                                                                                  |
|-------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Providing need-based support (equity)</b> | Help tailored to individuals.                        | Audio descriptions for visually impaired students; a screen reader so a blind student in Karachi can hear a digital textbook.      |
| <b>2. Removing barriers</b>                     | Help students in remote areas attend online classes. | Mobile-friendly, <b>low-data</b> apps so a student in Tharparkar or the katcha areas of Sindh can join without a fibre connection. |
| <b>3. Enabling active participation</b>         | Students who miss live class can still take part.    | <b>Recorded lectures</b> so load-shedding or an internet outage in Hyderabad does not make a student fall behind.                  |
| <b>4. Supporting different learning needs</b>   | Offer more than one format.                          | Videos, text notes and audio of the same Chapter 5 topic so each student learns in the way that suits them best.                   |

#### WORKED EXAMPLE

**Worked example: design four functions for one class WhatsApp + Classroom group in Nawabshah.**

- 1. Need-based support:** Ali is visually impaired — share materials as selectable text (not only photos) so a screen reader can speak them; add a short audio description of each diagram.
- 2. Remove barriers:** post a 5 MB audio summary instead of a 200 MB video for students on expensive mobile data.
- 3. Active participation:** upload the Zoom recording within an hour; students whose lights went out at 7 pm watch it at 11 pm.
- 4. Different learning needs:** the same lesson as (a) a 12-minute video, (b) a one-page PDF, (c) an Urdu audio note.

**Discussion from the source:** if a teacher **only** accepts assignments via an app that needs a **high-end smartphone**, is that equal access? **No**. It creates a **hardware barrier** for lower-income students. Web-based or light apps give better access.

#### REMEMBER IT

**Four functions: Support, Remove, Participate, Formats.** “Some Rural Pupils Fail” without them — Support (need-based), Remove barriers, Participate via recordings, Formats (video/text/audio).

#### COMMON MISTAKE

**Mistake:** Describing only captions and stopping.

**Fix:** Examiners want all four functions. Include remote-area low-data apps, recorded lectures for power cuts, and multiple formats — these are in the Sindh source on purpose.

#### اردو میں سمجھیں

انصاف اور مساوی رسائی کے چار کام ہیں: ضرورت کے مطابق مدد (جیسے نابینا طالب علم کے لیے آڈیو وضاحت)، رکاوٹیں ہٹانا (دور دراز علاقوں کے لیے کم ڈیٹا والی ایپس)، فعال شرکت ممکن بنانا (لوڈ شیڈنگ کے لیے ریکارڈ شدہ لیکچر)، اور مختلف اندازِ تعلم کی حمایت (ویڈیو، متن، آڈیو)۔ مہنگے فون والی ایپ پر صرف اسی طرح اسائنمنٹ لینا مساوی رسائی نہیں۔

#### EXAM TIP

Golden question: “Explain the functions of equity and equal access in digital collaboration” (5–8 marks). Write the four headings in order, one example each, and add the hardware-barrier point (high-end app vs light/web app). Another favourite: “Differentiate equal access and equity with examples” — same login vs captions.

#### CHECK YOURSELF

1. List the four functions of equity and equal access.
2. How do recorded lectures enable active participation in Pakistan?
3. Why is a high-end-smartphone-only assignment unfair?

Answers: 1. Need-based support; removing barriers; enabling active participation; supporting different learning needs. 2. Students who face power or internet outages can watch later and not fall behind. 3. It creates a hardware barrier for lower-income students; web-based or light apps give better access.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

### MCQ 1. Equity in digital collaboration means:

- a. Fair access to tools and participation, not only "same link for all"
- b. Giving everyone the identical laptop always without support
- c. Banning assistive tech
- d. English-only 4pt PDFs

### MCQ 2. Equal access includes:

- a. Devices, connectivity, language, disability support
- b. Only the top 10 students
- c. Only computer labs at midnight
- d. Only paid apps

### MCQ 3. A student on 2G with no laptop needs:

- a. Low-data options / lab time
- b. A 4K video lecture only
- c. A VR headset
- d. A GPU DL assignment first

### MCQ 4. Collaboration tools fail equity when:

- a. They require an expensive phone feature
- b. They have captions and Urdu
- c. They work on a lab PC
- d. They allow keyboard only

### MCQ 5. This is ★ for 5.5.1: functions of equity are:

- a. Inclusion, fairness of voice, accessible tools, support
- b. Faster CPUs
- c. Shorter OSI
- d. Nicer NAND

### MCQ 6. A fair group project:

- a. Roles + shared docs + offline option
- b. One member who owns the only laptop does all
- c. Voice chat only at 2am
- d. Uncaptioned 1-hour video mandatory

## Extra short questions

1. Define equity vs "everyone got the WhatsApp".
2. Functions of equity (book).
3. Two practical college moves.
4. Link to XI assistive tech.

## Extra long question

1. A group of 5: one blind, one with no smartphone, one in a village on weekends. Design the collaboration so all can submit. 10 marks ★.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Same message is not the same opportunity if some have no data bundle or cannot hear.
2. Access, participation, support, accessible design, respect.
3. Lab hours; printed alternative; captions; Urdu UI; extra time.
4. Screen readers + accessible docs are equity, not charity.

### Long

1. Shared doc that works in lab PCs; keyboard/screen reader; tasks that don't need 2am Zoom; recorded captions; warden-lab timetable. Assessment not based on who has 4G.